

From Grandpa's Workshop to Satellite Circuits: Ana Inês Inácio's Blueprint for the Future of Wireless

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Most of us never think about the radio waves crisscrossing the air around us. But for Ana Inês Inácio, those invisible signals are a canvas. At the Netherlands Organization for Applied Scientific Research (TNO) in The Hague, she designs **integrated circuits** the tiny, hyper-specialized chips that will form the backbone of tomorrow's radar and satellite communication systems. Her work recently earned her the IEEE-Eta Kappa Nu Outstanding Young Professional Award, a prestigious honor recognizing not just her technical prowess but her leadership in building global engineering communities. Inácio's path to RF engineering began far from high-tech labs, in the rural Portuguese village of Vales do Rio. Her grandfather repaired industrial looms in the region's textile mills, teaching himself electrical systems through correspondence courses. "He would show me why something broke and how we could fix it," she recalls. That hands-on curiosity stuck. In high school, she was torn between physics and biology, but encouragement from teachers and an engineer uncle steered her toward electronics. She enrolled in a five-year integrated master's program at the University of Aveiro, and a 2012 exchange at Eindhoven University of Technology brought her to the Netherlands, where she stayed. Today, her circuits are pushing the boundaries of **radio-frequency (RF) sensor systems**. These systems are crucial for advanced radar think of them as the eyes and ears of **autonomous vehicles** or climate-monitoring satellites. The award citation highlights her "**pioneering** advancements in RF **sensor systems**, bridging technical excellence with impactful community engagement." That community piece is no small add-on. Inácio has been a driving force in IEEE Young Professionals, creating spaces where early-career engineers can connect, collaborate, and innovate. "Sometimes that means circuits; sometimes it means helping people connect and grow together," she says. Her dual focus on atoms and on people offers a blueprint for how the next generation of technology leaders can build both better devices and a more **inclusive** field.

Word Treasure

integrated circuits -- Tiny chips that combine many electronic components on a single piece of semiconductor, used in almost all modern electronics.

radio-frequency -- A range of electromagnetic wave frequencies used for wireless communication, like radio, TV, and radar signals.

sensor systems -- Sets of devices that detect and respond to physical inputs - such as light, heat, or motion - and turn them into data.

autonomous vehicles -- Cars, drones, or other machines that can sense their environment and navigate without a human driver.

pioneering -- Being among the first to explore a new area of knowledge or to develop new methods.

inclusive -- Open to everyone, making sure that different people can participate and feel

they belong.

Background you need

Ana Inês Inácio works at the Netherlands Organization for Applied Scientific Research (TNO), an independent Dutch institute that develops technologies for industry and government, bridging science and real-world products.

The award she received, IEEE-Eta Kappa Nu Outstanding Young Professional, is given by a prestigious society within IEEE - the world's largest technical professional organization - to engineers under 35 who show both technical skill and community leadership.

Radio-frequency engineering is the backbone of modern life: every Wi-Fi router, smartphone, and weather satellite depends on circuits that send and receive these invisible signals. This field took off in the mid-20th century as radar and broadcasting expanded, and it continues to evolve with 5G and space exploration.

Inácio's path reflects a broader push for diversity in STEM fields, where women remain underrepresented. Her dual dedication to hardware and mentorship echoes trends seen in other pioneering engineers who believe technology advances fastest when everyone can contribute.

Break it down

LEAD: Invisible radio waves as a canvas; Ana Inês Inácio designs future backbone chips for radar and satellites.

+ WHO: Ana Inês Inácio, a Portuguese-born RF engineer at TNO in The Hague.

+ HONOR: She received the IEEE-Eta Kappa Nu Outstanding Young Professional Award.

BACKGROUND: Her journey from a rural Portuguese village to high-tech labs.

+ ROOTS: Grandfather repaired industrial looms in Vales do Rio, teaching her to fix things.

+ EDUCATION: Path from University of Aveiro to an exchange at Eindhoven and eventually settling in the Netherlands.

CURRENT WORK: Pushing the limits of RF sensor systems.

+ APPLICATION: Acting as the 'eyes and ears' for autonomous vehicles and climate-monitoring satellites.

DUAL FOCUS: Bridging technical excellence and community building.

+ TECH: Pioneering advancements in RF sensor systems, as noted in the award citation.

+ COMMUNITY: Driving IEEE Young Professionals to help early-career engineers connect and collaborate.

TAKEAWAY: Her blueprint shows that future technology leaders can build both better devices and a more inclusive field.

Why it matters

This story shows that the technology behind your smartphone and future self-driving cars is created by people who also invest in helping others succeed - challenging the idea that engineering is just about machines, not about human connections and inclusivity.

Test what you remember

1. What recent award did Ana Inês Inácio win?

- ☐ (A) IEEE Medal of Honor
- ☐ (B) Nobel Prize in Physics
- ☐ (C) IEEE-Eta Kappa Nu Outstanding Young Professional Award
- ☐ (D) European Inventor Award

2. Where did Inácio grow up?

- ☐ (A) The Hague, Netherlands
- ☐ (B) Vales do Rio, Portugal
- ☐ (C) Aveiro, Portugal
- ☐ (D) Eindhoven, Netherlands

3. What type of circuits does Inácio design at TNO?

- ☐ (A) Digital logic circuits for gaming consoles
- ☐ (B) Power circuits for household appliances
- ☐ (C) Integrated circuits for radar and satellite communication
- ☐ (D) Quantum circuits for computing

4. What systems are described as the 'eyes and ears' of autonomous vehicles in the article?

- ☐ (A) Camera and microphone systems
- ☐ (B) RF sensor systems
- ☐ (C) GPS navigation systems
- ☐ (D) Laser headlights

5. What helped spark Inácio's early interest in engineering?

- ☐ (A) A school robotics competition
- ☐ (B) Watching her grandfather repair industrial looms
- ☐ (C) A space shuttle launch on TV
- ☐ (D) Reading a biography of Marie Curie

6. What is the 'dual focus' Inácio is known for, according to the article?

- ☐ (A) Physics and biology
- ☐ (B) Teaching and designing
- ☐ (C) Atoms and people - technical work and community building
- ☐ (D) Radar and satellite systems

Answer key (try the quiz first!): 1: C 2: B 3: C 4: B 5: B 6: C

